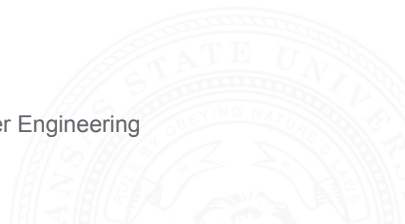


Line Parameters IV

Calculate one verified R-X-B record from conductor and tower data

Buxin She

Mike Wiegiers Department of Electrical and Computer Engineering



Lecture outline

Start with the tower: identify each circuit, its three phases, and the members of each phase bundle. Then build one R , X , B record per phase of each circuit.

- 1 Configuration: read the tower and distinguish phases, bundles, and circuits.
- 2 Resistance: use shared phase voltage and current division.
- 3 Inductance and capacitance: include shared magnetic and electric fields.
- 4 Route record: form physical totals, then convert to per unit.
- 5 Exercises: assemble and check a complete bundle record.

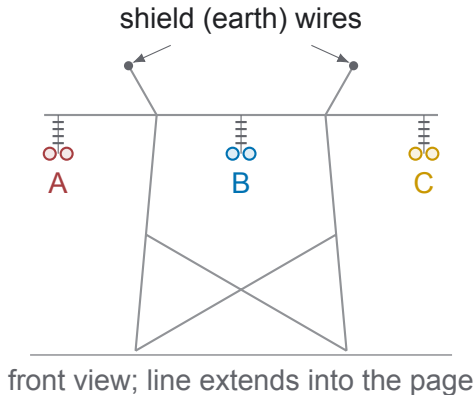
Produce one consistent R , X , B record for a bundled phase without confusing member count and circuit count.

PART 1

Tower, bundle configuration, and units

First read the physical tower. Identify the circuits, phases, and bundle members before extracting distances or combining conductor data.

Reading a transmission tower



Tower and crossarms

Support the conductors and maintain clearances.

Insulator strings

Support and electrically isolate the phase conductors.

Shield wires

Provide lightning protection; they are not extra phases.

Each pair: one phase, two subconductors.

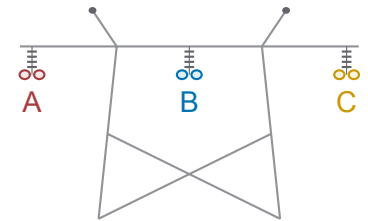
First identify A, B, and C. Then count the members within each phase.

Source: Course schematic, not to scale. Component roles: National Grid, "What is a pylon?"

One tower can carry two circuits

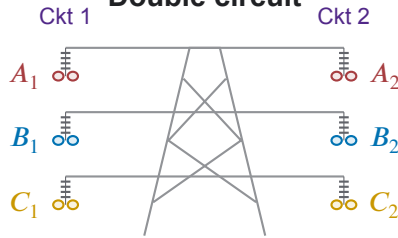
One three-phase circuit contains one A–B–C set. Each phase may be bundled.

Single circuit



$3 \times 2 = 6$ phase conductors

Double circuit



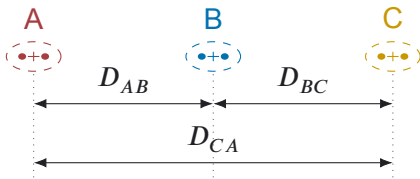
$2 \times 3 \times 2 = 12$ phase conductors

Illustrative layouts; shield wires excluded from counts and omitted on the right.

Trace the A–B–C set removed by opening circuit 1. A bundle member is only part of one phase.

From tower geometry to model inputs

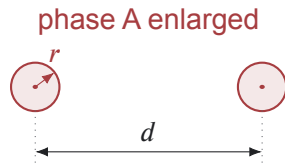
Keep the phase centers, then zoom into one bundle.



distances between phase-bundle centers

For complete transposition,

$$D_m = (D_{AB} D_{BC} D_{CA})^{1/3}.$$



$n_b = 2$ members in this phase

d : member spacing; r : outside radius.

D_s : magnetic GMR from conductor data.

Use phase spacing for GMD and member spacing for the bundle equivalent. Assume $d \ll$ phase spacing.

Common configurations within one phase

Each sketch is **one phase**, viewed in cross-section. Members share phase voltage.

Single conductor

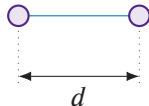
$$n_b = 1$$



no bundle spacing

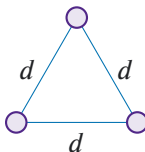
Two members

$$n_b = 2$$



Three members

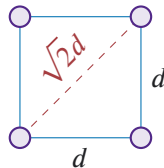
$$n_b = 3$$



equilateral triangle

Four members

$$n_b = 4$$



square

Center-to-center distances; d is the square side. Lines are geometric guides.

Square bundle: why are the three distances from one corner d , d , and $\sqrt{2}d$?

Source: Course redraw of common configurations: Glover et al., sixth edition (2017), Fig. 4.14.

Bundle input data

Each subconductor

- R'_{sub} : supplied AC resistance per length, at a stated temperature.
- D_s : supplied magnetic GMR.
- r : physical outside radius, half the outside diameter.

Geometry and operating data

- n_b : members per phase; d : side spacing for the symmetric bundles shown.
- D_{AB}, D_{BC}, D_{CA} : phase-center spacings.
- ℓ : route length; f : frequency.

Assume identical symmetric members and a completely transposed line. Use phase centers when the bundle is small relative to phase spacing.

Rates, totals, and per unit

Representation	Example	Meaning
Per length	R'_ϕ in Ω/km	one unit of a uniform phase
Physical total	R_ϕ in Ω	one phase over the full route
Per unit	r_{pu} in p.u.	total divided by a declared base

Distances inside a logarithm must use the same unit. A rate and its route length must also use matching length units.

Per unit length and per unit are different operations. Calculate the physical rate before either route scaling or base conversion.

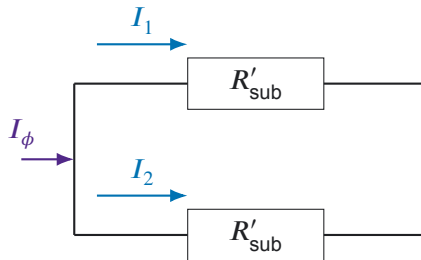
PART 2

Bundle resistance

The members share one phase voltage. Their current division determines the equivalent resistance seen by the total phase current.

Resistance per unit length

Identical symmetric members share the same terminal voltage.



resistive model of a unit-length slice

$$I_\phi = I_1 + I_2, \quad I_1 = I_2 = I_\phi/2.$$

The common resistive drop per length is

$$R'_{\text{sub}} \frac{I_\phi}{2} = R'_\phi I_\phi.$$

$$R'_\phi = R'_{\text{sub}}/2$$

For n_b identical, equally sharing members,
 $R'_\phi = R'_{\text{sub}}/n_b$.

Predict: if the same phase current is shared by two members, how does the resistive drop change?

PART 3

Bundle inductance and capacitance

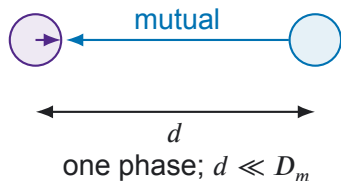
The resistance equivalent is known. Magnetic linkage and electric potential also include the other members, so their equivalents require field contributions.

Magnetic bundle equivalent

self term: D_s

$$I_1 = I_\phi/2$$

$$I_2 = I_\phi/2$$



$$L'_\phi = \frac{\lambda'_1}{I_\phi} = 2 \times 10^{-7} \ln \frac{D_m}{D_{sL}},$$

Both currents link member 1:

$$\lambda'_1 \approx 2 \times 10^{-7} \left[I_1 \ln \frac{D_m}{D_s} + I_2 \ln \frac{D_m}{d} \right].$$

Equal currents give the mean of two logarithms:

$$\frac{1}{2} \left[\ln \frac{D_m}{D_s} + \ln \frac{D_m}{d} \right] = \ln \frac{D_m}{\sqrt{D_s d}}.$$

$$D_{sL} = \sqrt{D_s d}.$$

Identify the term that would be lost by treating the two inductances as independent parallel elements.

Source: Glover et al., sixth edition (2017), Sec. 4.6.

Inductance and reactance rates

Completely transposed phase geometry

$$D_m = (D_{AB}D_{BC}D_{CA})^{1/3}$$

Using the magnetic bundle self-distance,

$$L'_\phi = 2 \times 10^{-7} \ln \frac{D_m}{D_{sL}} \quad [\text{H/m}]$$

$$X'_\phi = 2\pi f L'_\phi \quad [\Omega/\text{m}]$$

Larger bundle self-distance reduces the logarithmic ratio and the inductance rate.

Calculate L'_ϕ from geometry, then X'_ϕ from frequency. Route length enters later.

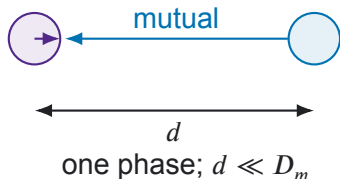
Source: Glover et al., sixth edition (2017), Sec. 4.6.

Electric bundle equivalent

self term: r

$$q'_1 = q'_\phi/2$$

$$q'_2 = q'_\phi/2$$



$$D_{sC} = \sqrt{rd},$$

Common phase voltage:

$$V_{\phi n} \approx \frac{1}{2\pi\epsilon} \left[q'_1 \ln \frac{D_m}{r} + q'_2 \ln \frac{D_m}{d} \right].$$

With equal charge sharing,

$$V_{\phi n} = \frac{q'_\phi}{2\pi\epsilon} \ln \frac{D_m}{\sqrt{rd}}.$$

$$C'_\phi = \frac{q'_\phi}{V_{\phi n}} = \frac{2\pi\epsilon}{\ln(D_m/D_{sC})}.$$

Which self distance belongs in the electric term: physical radius r or magnetic GMR D_s ? Explain from the conductor surface.

Source: Glover et al., sixth edition (2017), Sec. 4.10.

Same averaging, different bundle geometry

Identical symmetric members: equal sharing again gives a geometric mean.

One phase	Magnetic GMR D_{sL}	Electric radius D_{sC}
Single conductor	D_s	r
Two members	$(D_s d)^{1/2}$	$(rd)^{1/2}$
Three (equilateral)	$(D_s d^2)^{1/3}$	$(rd^2)^{1/3}$
Four (square)	$(\sqrt{2} D_s d^3)^{1/4}$	$(\sqrt{2} rd^3)^{1/4}$

For one square corner:

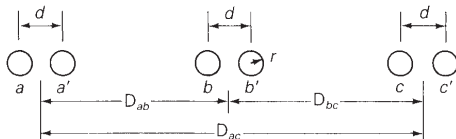
$$D_{sL} = (D_s \cdot d \cdot d \cdot \sqrt{2}d)^{1/4} = 2^{1/8} (D_s d^3)^{1/4}.$$

$2^{1/8} \approx 1.091$: the diagonal explains the coefficient; $D_s \rightarrow r$ gives the electric equivalent.

Use the same L' and C' formulas with the appropriate self-distance. Phase-center GMD requires bundle size $\ll$ phase spacing.

Source: Glover et al., sixth edition (2017), Eqs. 4.6.19–4.6.21 and 4.10.5–4.10.7.

Capacitance and susceptance rates



Two members form one phase at a common potential. Ground effects are neglected here.

Air dielectric

$$C'_n = \frac{2\pi\epsilon_0}{\ln(D_m/D_{sC})} \quad [\text{F/m}]$$

$$B'_{\text{sh}} = 2\pi f C'_n \quad [\text{S/m}]$$

$$\epsilon_0 \approx 8.85419 \times 10^{-12} \text{ F/m.}$$

The electric equivalent determines C'_n . Frequency gives B'_{sh} ; length has not yet entered.

Source: Glover et al., sixth edition (2017), Fig. 4.22 and Sec. 4.10.

PART 4

Route totals and per unit

Bundle geometry now supplies per-length parameters. A network model needs full-route totals on its declared system bases.

Physical route totals

Series parameters of one phase

$$R_{\phi} = R'_{\phi} \ell, \quad X_{\phi} = X'_{\phi} \ell$$

$$Z = R_{\phi} + jX_{\phi} \quad [\Omega]$$

Shunt parameters of one phase

$$C_n = C'_n \ell, \quad B_{\text{sh}} = B'_{\text{sh}} \ell$$

$$Y = jB_{\text{sh}} \quad [\text{S}]$$

These are full-route totals for one phase of one physical circuit. They contain the bundle effect already.

Multiply by the compatible route length once. Do not multiply the branch parameters by three phases.

Per-unit line record

The physical totals are ready. To join equipment from the same voltage zone in one network model, choose line-to-line voltage base $V_{LL,base}$ and total three-phase power base S_{base} .

$$Z_{base} = \frac{V_{LL,base}^2}{S_{base}}, \quad Y_{base} = \frac{1}{Z_{base}}$$

Series normalization

$$r_{pu} = \frac{R_{\phi}}{Z_{base}}, \quad x_{pu} = \frac{X_{\phi}}{Z_{base}}$$

Shunt normalization

$$b_{pu} = \frac{B_{sh}}{Y_{base}} = B_{sh} Z_{base}$$

Normalize the complete physical route totals. Per-unit values are dimensionless and require stated bases.

Charging quantities

For a charging estimate at uniform line-to-line RMS voltage V_{LL} ,

$$V_{\phi} = \frac{V_{LL}}{\sqrt{3}}, \quad |I_{\text{ch},\phi}| = B_{\text{sh}} V_{\phi}$$

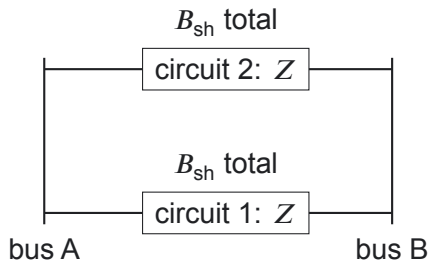
$$Q_C = 3 B_{\text{sh}} V_{\phi}^2 = B_{\text{sh}} V_{LL}^2$$

$Q_C > 0$ is capacitive reactive power supplied. Signed absorbed power is $-Q_C$.

A nominal- π places $B_{\text{sh}}/2$ at each end. Different terminal voltages require separate endpoint calculations.

Charging uses total susceptance and operating voltage. The line's MVA limit remains separately supplied data.

Two parallel circuits



Identical circuits connect the same buses; mutual coupling is neglected.

$$Z_{\text{eq}} = \left(\frac{1}{Z} + \frac{1}{Z} \right)^{-1} = Z/2.$$

$$B_{\text{eq}} = B_{\text{sh}} + B_{\text{sh}} = 2B_{\text{sh}}.$$

Each circuit's Z , B_{sh} already includes its bundle. Keep both physical circuit records.

Predict: after one circuit opens, which parameters remain? Does changing the number of bundle members describe that outage?

Source: Glover et al., sixth edition (2017), Sec. 4.13 discusses mutual coupling.

PART 5

Exercises

Combine the member data, bundle geometry, route length, and bases in one calculation; keep the physical circuit identity attached.

Exercise 1: given and find

Given and find

Each phase has two identical members, spaced 0.50 m apart. Each member has GMR 0.0100 m, physical radius 0.0120 m, and resistance 0.0600 Ω/km .

- 1 Find bundle GMR, electric equivalent radius, and bundle resistance.
- 2 For a 100-km route, find its resistance per phase.
- 3 If two identical uncoupled circuits run in parallel, state the equivalent Z and B .

Exercise 1: solution

$$D_{sL} = \sqrt{0.0100(0.50)} = 0.07071 \text{ m}, \quad D_{sC} = \sqrt{0.0120(0.50)} = 0.07746 \text{ m}$$

$$R'_{\phi} = \frac{0.0600}{2} = 0.0300 \text{ } \Omega/\text{km}, \quad R_{\phi} = 3.00 \text{ } \Omega$$

$$Z_{\text{parallel}} = Z_{\text{circuit}}/2, \quad B_{\text{parallel}} = 2B_{\text{circuit}}$$

The parallel equivalent assumes identical uncoupled circuits. Keep both physical circuit rows for an outage study.

Exercise 2: given

n_b is the number of bundle members per phase.

System and route

V_{LL}	$V_{LL,base}$	500.0 kV
f		60.0 Hz
ℓ		80.0 mi = 128.74752 km
S_{base}		100.0 MVA

Balanced, fully transposed, one physical circuit. Mutual distances use phase-bundle centers because $d \ll$ phase spacing.

Supplied planning input: 1000 MVA/circuit, independent of the $R-X-B$ derivation.

Tower and conductor

D_{AB}, D_{BC}, D_{CA}	32, 32, 64 ft
n_b, d	2, 1.50 ft
$R'_{sub,ac,25^\circ C}$	0.0105 Ω/kft
D_s	0.0588 ft
outside diameter	1.762 in

All Exercise 2 inputs are stated here. The supplied planning limit remains separate from the parameter derivation.

Exercise 2: find

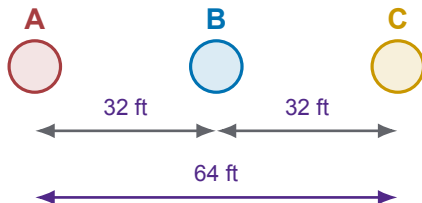
Find and verify

- (a) Find phase GMD and the two-member resistance rate.
- (b) Find magnetic and electric self-distances, then L'_{ϕ} , X'_{ϕ} , C'_n , and B'_{sh} .
- (c) Form the full 80-mi physical totals, charging current, and supplied Mvar.
- (d) Convert to the 500-kV, 100-MVA bases. Compare one member with the two-member bundle and verify the result.

State the units and model assumptions with each result.

Ex. 2(a): phase GMD

Small-bundle phase-center approximation



$$\begin{aligned} D_m &= (D_{AB} D_{BC} D_{CA})^{1/3} \\ &= (32 \cdot 32 \cdot 64)^{1/3} \\ &\approx \boxed{40.3175 \text{ ft}}. \end{aligned}$$

Since $d = 1.50 \text{ ft} \ll 32 \text{ ft}$, each mutual bundle distance is represented by its phase-center spacing.

Complete transposition replaces the three phase spacings by their geometric mean.

Ex. 2(a): resistance rate

Supplied 25°C AC rate

$$\begin{aligned} R'_\phi &= \frac{0.0105}{2} \Omega/\text{kft} \\ &= 0.00525 \Omega/\text{kft}. \end{aligned}$$

Since 1 mi = 5.28 kft,

$$R'_\phi = (0.00525)(5.28) = 0.02772 \Omega/\text{mi}.$$

Equivalent metric rate:

$$R'_\phi \approx 0.0172244094 \Omega/\text{km}.$$

The derived bundle rate retains the supplied 25°C AC condition.

Resistance divides by two for the bundle, then the length unit converts. The number of phases never multiplies this rate.

Ex. 2(b): magnetic self-distance

Use one member's supplied GMR and the member spacing:

$$D_{sL} = \sqrt{D_s d} = \sqrt{(0.0588)(1.50)} = 0.296985 \text{ ft}$$

Both phase GMD and bundle self-distance are in feet:

$$\frac{D_m}{D_{sL}} = \frac{40.3175}{0.296985}$$

$$\ln(D_m/D_{sL}) = 4.91085912$$

This dimensionless magnetic ratio determines the inductance rate.

Ex. 2(b): inductance and reactance

$$L'_{\phi} = 2 \times 10^{-7} (4.91086) \approx 9.821718242 \times 10^{-7} \text{ H/m.}$$

Over one mile,

$$L'_{\phi} \approx 1.580652332 \text{ mH/mi.}$$

At 60 Hz,

$$X'_{\phi} = 2\pi(60)(1.58065 \times 10^{-3}) \approx 0.595891891 \text{ } \Omega/\text{mi.}$$

The reactance rate is already a balanced per-phase quantity and enters the circuit row directly.

Ex. 2(b): electric self-distance

The supplied 1.762-in outside dimension is a diameter. Convert it to a radius in feet:

$$r = \frac{1.762}{2} \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}} = 0.0734167 \text{ ft}$$

$$D_{sC} = \sqrt{rd} = \sqrt{(0.0734167)(1.50)} = 0.331851 \text{ ft}$$

$$\ln(D_m/D_{sC}) = 4.79985456$$

Electric self-distance uses physical radius. It differs from the magnetic self-distance.

Ex. 2(b): capacitance and B rate

$$\begin{aligned}C'_n &= \frac{2\pi(8.8541878 \times 10^{-12})}{4.79985456} \text{ F/m} \\&= 11.5904559 \text{ nF/km} \\&= 18.6530307 \text{ nF/mi.}\end{aligned}$$

At 60 Hz,

$$\begin{aligned}B'_{\text{sh}} &= 2\pi(60)(18.6530307 \times 10^{-9}) \text{ S/mi} \\&= 7.03203 \text{ }\mu\text{S/mi.}\end{aligned}$$

These are per-mile rates. Apply the 80-mi route length on the next page.

Ex. 2(c): physical totals

Quantity	Per-mile rate	One-circuit route total
R_ϕ	0.02772 Ω/mi	2.2176 Ω
X_ϕ	0.595892 Ω/mi	47.6714 Ω
C_n	18.653 nF/mi	1.49224 μF
B_{sh}	7.03203 $\mu\text{S}/\text{mi}$	0.562562 mS

$$Z = 2.2176 + j47.6714 \Omega, \quad Y = +j0.562562 \text{ mS}.$$

Every total belongs to one phase of one physical circuit over the full 80-mile route.

Ex. 2(c): charging quantities

The balanced shunt uses phase-to-neutral voltage:

$$V_{\phi,\text{rms}} = \frac{500 \text{ kV}}{\sqrt{3}} \approx 288.675135 \text{ kV.}$$

$$|I_{\text{ch},\phi}| = B_{\text{sh}} V_{\phi,\text{rms}} \approx \boxed{162.398 \text{ A.}}$$

The three-phase charging magnitude closes directly:

$$\boxed{|Q_{\text{ch},3\phi}| = B_{\text{sh}} V_{LL,\text{rms}}^2 \approx 140.640538 \text{ Mvar.}}$$

The line rating remains a supplied planning limit. Charging Mvar is a separate electrical result from B_{sh} and voltage.

Ex. 2(d): per-unit record

Declared bases

$$Z_{\text{base}} = \frac{(500 \text{ kV})^2}{100 \text{ MVA}} = 2500 \, \Omega,$$

$$Y_{\text{base}} = 0.0004 \text{ S} = 0.400 \text{ mS}.$$

Course worksheet record

$$r_{\text{pu}} = 0.00088704,$$

$$x_{\text{pu}} = 0.0190685,$$

$$b_{\text{pu}} = 1.40641.$$

b_{pu} is the total one-circuit shunt-susceptance magnitude. The supplied planning limit is 1000 MVA.

The physical line-data record is complete. Software-specific mapping remains a later task.

Ex. 2(d): bundle comparison

Parameter comparison under one fixed geometry

The phase centers, subconductor type, route, frequency, and model remain fixed while bundle count changes. Material-quantity and rating comparisons require a separate design study.

Rate	One member	Two-member bundle	Change
R'_ϕ [Ω/mi]	0.05544	0.02772	-50.000%
X'_ϕ [Ω/mi]	0.79241	0.595892	-24.800%
C'_n [nF/km]	8.81881	11.5905	+31.429%

Under the declared assumptions, bundling lowers resistance and reactance while increasing capacitance and charging.

Ex. 2(d): verification

Check	Expected signature
Self quantities	D_{sL} appears in X , and D_{sC} appears in C .
Route-length closure	Per-mile rate times 80 mi returns each unprimed total exactly once.
Charging closure	$B_{sh} V_{LL,rms}^2 = 140.641 \text{ Mvar}$.
Per-unit closure	$b_{pu} S_{base} = 140.641 \text{ Mvar at } 1.0 \text{ p.u.}$
Circuit identity	Each physical circuit keeps a full row, ID, and separately supplied limit.

A correct answer must close numerically, dimensionally, physically, and topologically.

Summary

- **Bundle self quantities:** magnetic calculations use bundle GMR. Capacitance uses an equivalent radius based on physical conductor radius.
- **Complete line record:** resistance, reactance, and total shunt susceptance must share the same geometry, length, and circuit identity.
- **Per-unit conversion:** convert the complete physical row with the correct system power base and voltage-zone base.
- **Parallel circuits:** retain separate rows for separate physical circuits. The simple $Z/2$, $2B$ equivalent assumes identical uncoupled circuits; nearby circuits can require mutual terms.

A reliable branch model preserves both its electromagnetic data and its topological identity.